

Problem 1:

Let R be a noetherian ring such that $\text{Spec}(R)$ is finite. Prove that $\dim R \leq 1$, and give an example for which $\dim R = 1$.

Proof.

Assume R is noetherian and that $\text{Spec}(R)$ is finite.

Let $\mathfrak{p} \in \text{Spec}(R)$. Let $\{q_1, \dots, q_n\}$ be the primes in $\text{Spec}(R)$ that do not contain $\mathfrak{p}$. This is a finite list since $\text{Spec}(R)$ is finite.

By the prime avoidance lemma this implies that $\mathfrak{p} \not\subset \bigcup_{i=1}^n q_i$.

This implies there exists an $x \in \mathfrak{p} \setminus \bigcup_{i=1}^n q_i$.

If $\tau \not\supset \mathfrak{p}$ and $\tau \in \text{Spec}(R)$ this implies that $\tau = q_i$ for some i . Thus, $x \notin \tau$.

If $\tau \supset \mathfrak{p}$ and $\tau \in \text{Spec}(R)$ this implies that $x \in \tau$.

Thus, $x \in \tau \iff \mathfrak{p} \subset \tau$.

This implies $V(x) = \{q \in \text{Spec}(R) \mid x \in q\} = \{q \in \text{Spec}(R) \mid \mathfrak{p} \subset q\} = V(\mathfrak{p})$.

This implies that $\sqrt{(x)} = \sqrt{\mathfrak{p}} = \mathfrak{p}$.

Thus, $\mathfrak{p}$ is minimal over (x) .

Therefore, by the principle ideal theorem this implies that $\dim \mathfrak{p} \leq 1$.

Thus, since $\mathfrak{p}$ is arbitrary we get that $\dim R \leq 1$.

Example $\mathbb{Z}_{(p)}$ s.t. p is prime. The prime ideals in here are $(0), (p)R_{(p)}$. Thus, $\dim \mathbb{Z}_{(p)} = 1$. ■

Problem 2:

Let R be a noetherian ring, and let $R[x]$ be the polynomial ring over R in one variable. Let $R[x, x^{-1}]$ be the localization of $R[x]$ at the element x . Prove that $\dim R[x, x^{-1}] = \dim R + 1$.

Proof.

Assume R is noetherian.

Shown in class $\dim R[x] = \dim R + 1$. Also, shown in class since $R[x, x^{-1}]$ is a localization of $R[x]$ this implies $\dim R[x, x^{-1}] \leq \dim R[x] = \dim R + 1$.

First if $\dim R = \infty$ then we can extend a chain to $R[x]$ by just changing $\mathfrak{p}$ with $\mathfrak{p}[x]$ which was proven prime in class. Then extending $R[x, x^{-1}]$ since $x \notin \mathfrak{p}[x]$ for any prime $\mathfrak{p}$ this implies that this is also a chain in $R[x, x^{-1}]$ since primes in $R[x]$ s.t. x is not in the prime are in bijection with primes in $R[x, x^{-1}]$.

Thus, $\dim R[x, x^{-1}] = \infty = \dim R + 1$

If $\dim R$ is finite.

Let $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ be a max chain of prime ideals in R where $n = \dim R$.

First we can extend this to $\mathfrak{p}_0[x] \subsetneq \mathfrak{p}_1[x] \subsetneq \cdots \subsetneq \mathfrak{p}_n[x]$.

Also consider $\mathfrak{p}_n[x] + (x + 1)$ this is also prime since $R[x]/(\mathfrak{p}_n[x] + (x + 1)) \cong R/\mathfrak{p}_n$.

Thus, $\mathfrak{p}_0[x] \subsetneq \mathfrak{p}_1[x] \subsetneq \cdots \subsetneq \mathfrak{p}_n[x] \subsetneq \mathfrak{p}_n[x] + (x + 1)$ is a chain. Also, since x is not in any of these prime ideals.

This implies that this is also a chain in $R[x, x^{-1}]$. Thus, $\dim R[x, x^{-1}] \geq \dim R + 1$.

$\dim R[x, x^{-1}] = \dim R + 1$. ■

Problem 3:

Let R be a noetherian ring.

(a)

Show that if $\mathfrak{p} \subset R$ is a prime ideal, then for any $n \geq 1$ its n th symbolic power $\mathfrak{p}^{(n)}$ is the smallest $\mathfrak{p}$ -primary ideal containing $\mathfrak{p}^n$. That is, if $\mathfrak{q}$ is another $\mathfrak{p}$ -primary ideal containing $\mathfrak{p}^n$, then $\mathfrak{p}^{(n)} \subset \mathfrak{q}$.

Proof.

Let $\mathfrak{p}$ be a prime ideal of R .

Assume $\mathfrak{q}$ is $\mathfrak{p}$ -primary ideal s.t. $\mathfrak{p}^n \subset \mathfrak{q}$.

Let $x \in \mathfrak{p}^{(n)} \implies \exists s \notin \mathfrak{p}$ s.t. $sx \in \mathfrak{p}^n \subset \mathfrak{q}$.

Thus, $xs \in \mathfrak{q}$. Since $\mathfrak{q}$ is $\mathfrak{p}$ -primary this implies either $x \in \mathfrak{q}$ or $s^n \in \mathfrak{p}$, but if $s^n \in \mathfrak{p} \implies s \in \mathfrak{p}$ which is a contradiction.

Therefore, $x \in \mathfrak{q}$. Thus, $\mathfrak{p}^{(n)} \subset \mathfrak{q}$. ■

(b)

Show that if $\mathfrak{m} \subset R$ is a maximal ideal, then $\mathfrak{m}^n = \mathfrak{m}^{(n)}$ for all $n \geq 1$

Proof.

Let $\mathfrak{m}$ is a maximal ideal.

Consider $\mathfrak{m}^n$. First $\mathfrak{m}^n \subset \mathfrak{m}^{(n)}$

Since $\sqrt{\mathfrak{m}^n} = \mathfrak{m}$ is maximal this implies that $\mathfrak{m}^n$ is $\mathfrak{m}$ -primary.

Thus, by (a) this implies that $\mathfrak{m}^{(n)} \subset \mathfrak{m}^n$.

Therefore, $\mathfrak{m}^{(n)} = \mathfrak{m}^n$ ■

Problem 4:

Let R be a noetherian ring and $f \in R$.

(a)

Prove that if f is non-zero-divisor in R , then $\dim R/(f) \leq \dim R - 1$.

Proof.

Assume f is a non-zero-divisor.

Assume for contradiction that $f \in \mathfrak{p}$ s.t. $\mathfrak{p}$ is a minimal prime.

Thus, consider $R_{\mathfrak{p}}$. Since there is only one maximal ideal in the localization this implies it is artinian.

Thus, $\mathfrak{p}R_{\mathfrak{p}}^n = 0$ for some $n \in \mathbb{N}$. This implies that $f^n/1 = 0$. Thus, there exists an $s \notin \mathfrak{p}$ s.t. $sf^n = 0$. Since f is a non-zero divisor this implies $s = 0$. This is a contradiction though.

Thus, f is not in any minimal prime.

Therefore, the length of chains of primes that contain f is at most length $\dim R - 1$ since f is not in any minimal primes.

Therefore, since there is a bijection between primes in R that contain f and primes in $R/(f)$. This implies that $\dim R/(f) \leq \dim R - 1$. ■

(b)

If $(R, \mathfrak{m})$ is a local ring and $f \in \mathfrak{m}$, prove that $\dim R - 1 \leq \dim R/(f) \leq \dim R$. In particular, if f is also a non-zero-divisor, then by the previous part we have $\dim R/(f) = \dim R - 1$.

Proof.

Assume $(R, \mathfrak{m})$ is a local ring and $f \in \mathfrak{m}$.

Let $\mathfrak{p}$ be a minimal prime over f .

By the principle ideal theorem this implies that $\text{codim}(\mathfrak{p}) \leq 1$.

This implies a max length chain that starts at $(\mathfrak{p})$ is $\text{codim}(\mathfrak{m}) - \text{codim}(\mathfrak{p}) \geq \dim R - 1$.

Thus, $\dim R/(f) \geq \dim R - 1$. ■

(c)

Give an example of a local ring $(R, \mathfrak{m})$ and $f \in \mathfrak{m}$ such that $\dim R - 1 = \dim R/(f)$, and give an example so that $\dim R = \dim R/(f)$.

Proof.

- Consider $k[[x]]$ formal power series of a field k in one variable. Proven in 494 $f \in k[[x]]$ is a unit if and only if $a_0 \neq 0$. Thus, the set of non units is exactly (x) . A proper ideal in $k[[x]]$ cannot contain a unit, thus every proper ideal is contained in (x) . Thus, (x) is the unique maximal ideal.

Let $f \in k[[x]]$ be non zero. Let a_n be the first term that is non zero in f . This implies that $f = t^n \cdot u$ s.t. u is a unit since its first term is non-zero which implies it is a unit.

Let $I \neq 0$ be an ideal. Let $f \in I$ s.t. $f = x^n \cdot u$ where n is minimal amount all elements in I . This comes from the well ordering principle since these powers are bounded below by zero and all natural numbers.

Thus, $x^n = f \cdot u^{-1} \in I$. Thus, $(x^n) \subset I$. Let $g \in I$. Write $g = x^m \cdot v$. Since n is minimal we have that $n \leq m$, so $g \in (x^n)$.

Thus, $I = (x^n)$. Thus, every non-zero ideal in $k[[x]]$ is of the form (x^n) for $n \geq 0$.

Lets consider prime ideals now

Since $k[[x]]$ is a domain (0) is prime.

Now let $\mathfrak{p} \neq 0$ be prime. Let $0 \neq f \in \mathfrak{p}$. As above we have that $f = x^n u$ since u is a unit this implies $x^n \in \mathfrak{p} \implies x \in \mathfrak{p}$. Thus, $x \in \mathfrak{p}$, so $\mathfrak{p} = (x)$. Therefore, the only prime ideals are (0) and (x) . Thus, the max chain is $(0) \subsetneq (x)$. Thus, $k[[x]]$ is dimension one.

Also, $k[[x]]/(x) \cong k$ which is dimension zero. Thus, $\dim k[[x]] - 1 = \dim k[[x]]/(x)$.

- Consider $R = \mathbb{R}$ then since (0) is the only prime ideal thus maximal we get that $\dim \mathbb{R} = 0 = \dim \mathbb{R}/(0)$



Problem 5:**Proof.**

Since $A = k[x]_{(x)}$ this implies that $\dim A \leq \dim k[x] = 1$, but since we have the chain $(0) \subsetneq (x)$ in A this implies $\dim A = 1$.

Let $R = A[t]$. Since A is noetherian this implies that $\dim R = \dim A[t] = \dim A + 1 = 2$.

Consider $A[t]/(xt - 1) \cong A[1/x]$.

Since $A = k[x]_{(x)}$. This implies we inverted everything that has a non zero constant term. Then in $A[1/x]$ we then localize again and invert x which then inverts non-zero polynomials with zero in the constant term.

Thus, we have inverted $k[x] \setminus 0$. Therefore, $A[1/x] \cong k(x)$. Therefore, $(xt - 1)$ is a maximal ideal.

Thus, $\dim R/\mathfrak{p} = 0$.

Also, since R is noetherian by the principle ideal theorem we get that $\text{codim}(\mathfrak{p}) \leq 1$.

Since R is a domain this implies that $(0) \subset \mathfrak{p}$. Thus, $\mathfrak{p}$ has codimension 1. ■